

SECTION C

11 (**long**) Consider a linear system with input $f(t)$ governed by the equation

$$\beta \frac{dy}{dt} + y = f(t)$$

where β is a positive constant.

(a) Find the step response of the system without using Laplace transforms, taking $y(t) = 0$ for $t < 0$. [5]

(b) Find the impulse response, $g(t)$, of the system. [4]

(c) Explain clearly why the response to a forcing function $f(t)$ can be written as

$$y(t) = \int_0^t f(\tau) g(t - \tau) d\tau$$

State any assumptions you make. [6]

(d) Find the system response for

$$f(t) = \begin{cases} 0 & t < 0 \\ \frac{1}{\alpha} \exp(-t/\alpha) & t \geq 0 \end{cases}$$

where $\alpha \neq \beta$ is a positive constant. What is the significance of the limit $\alpha \rightarrow 0$? Comment briefly on your result. [15]